

Genotype data



Quality Control

	A ₁	A ₂	A ₃	A ₄	A ₅	A ₆
V ₁ P ₁	1	0	0	0	0	0
V ₂ P ₁	0	0	1	0	0	1
V ₃ P ₁	1	0	2	0	0	0
V ₄ P ₁	0	0	1	0	0	0
V ₅ P ₁	1	1	0	0	0	2

Pre-filtering

Filtering variants (missing > 5%)
and samples (missing > 20%)



HRC Reference Panel Alignment

Strand Alignment
Allele Harmonization
Position Liftover

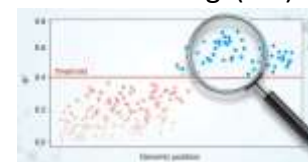
Phasing and Imputation



- Each Chromosome
- Eagle 2.4
- TOPMED r3 panel

Post Quality Control

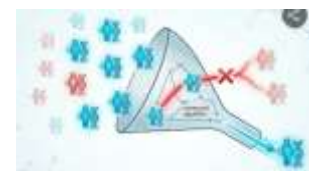
R2 filtering (0.4)



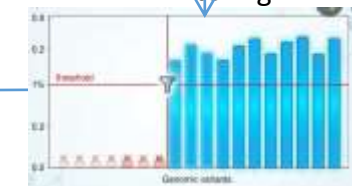
Annotation



Excluding duplicants

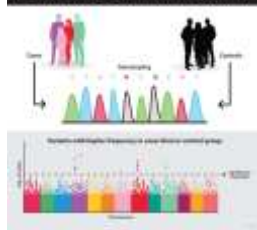


MAF filtering



Feature Selection

Genome-Wide Association Study (GWAS)



- Jansen et al (2019)*
- p-value threshold : $1e-4$
; $1e-6$; $1e-8$

rs11503252_A	rs4844000_A	rs12027841_T	rs11162438_G	rs11110662_T	...
0	1	0	2	1	...
0	1	0	1	1	...
0	1	0	2	0	...
...	...	...	...	...	...
0	1	0	2	1	...